Capacity Meters That Change Sign: Diagnosing Zero-Cost NAS Proxy Transport Across Search Spaces
Supplementary Material

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A. Notation

Table 1 collects the symbols used throughout.

Symbol	Meaning
Q	a zero-cost proxy (scalar per architecture)
acc	trained test accuracy (from the benchmark)
T	early-training descent rate (T_{slope})
erank	effective rank of the at-init feature spectrum
params	parameter count
cap	capacity controls {erank, params}
ρ, corr	Spearman rank correlation (empirical)
Corr	Pearson correlation (Prop. 1)
partial$(\cdot,\cdot \mid \cdot)$	rank partial correlation
R^2_{res}	out-of-fold R^2 on the capacity residual
$\mathcal{G}$	sub-σ-algebra of at-init (label-free) info
S	a $\mathcal{G}$-measurable (label-free) statistic
Y	frozen at-init NTK kernel-regression error
R^2_{cap}	frozen-kernel resol. limit
PC1	leading principal component of the oriented zoo
R_α	Rényi-α effective rank (α=1: NEAR)

Table 1. Notation.

B. Per-Conclusion Dependency Map

The body states results top-down; this table maps each load-bearing conclusion to its body location, key exhibit, and supporting appendix section.

Conclusion	Body	Key fig./tab.	App. evidence
Frozen-NTK bound	Sec. 3	Eq. (3)	Appendices F and G.5
Capacity factor	Sec. 4	Fig. 2a	Appendix G.6
Early-T association	Sec. 4	Tab. 1	Appendix G.4
At-init predictability	Sec. 4	Fig. 2	Appendix G.7
No empirical ceiling	Sec. 4	Fig. 2c	Appendix G.5
Raw sign reversals	Sec. 5	Fig. 1	Appendix G.1
Dimension sensitivity	Sec. 5	Tab. 12	Appendix G.2
Broader portability	Sec. 5	Fig. 3	Appendices G.3 and G.4
Search trap costs	Sec. 5	Tab. 3	Appendix I.4

Table 2. **Per-conclusion dependency map.** Each load-bearing conclusion, its body location, its key exhibit, and the appendix section carrying its full evidence.

C. The Proxy Zoo, Target T, and Sign Orientation

This section fixes the three objects the body leans on but does not re-derive: what a zero-cost proxy is, the kernel picture that *motivates* (but does not constrain) the decomposition, and the external trainability target T; it then records the per-axis sign-orientation rule that every aggregate analysis applies.

Zero-cost proxies. A zero-cost proxy scores an untrained network from a single forward/backward pass and is used to rank candidates during search without training [1]. The thirteen standard proxies we use are those collected in NAS-Bench-Suite-Zero [10]; our spectral arm (cond-sum, stable-sum, R_α) reimplements the Dextr [2] and NEAR [7] functionals, parity-checked in Appendix H.2.

The kernel spectral-bias picture. For kernel regression / infinitely wide networks, the generalization error decomposes over the kernel's Mercer spectrum $\{\rho_i\}$ and the target coefficients $\{\bar{c}_i\}$ (Appendix D). It cleanly separates a label-free *capacity* factor (the spectrum) from a label-dependent *alignment* factor (the coefficients), and motivates, without quantitatively constraining, our decomposition. We import only this qualitative split; the lazy/NTK regime it assumes need not hold at NB201 scale, consistent with an observed capacity-orthogonal association with early training (Sec. 4).

The trainability target T. T is the early-training learning rate read off the *real* NATS-Bench training curves (hp=12, epochs 0–4): the ordinary-least-squares slope of the first five train-accuracy points (larger T = faster early learning).

It is an external measurement (training curves, not proxies), so partialling a proxy against T compares against an outside quantity rather than performing a self-consistency check.

Sign orientation (per axis). Proxies disagree on which direction predicts higher accuracy, so before any aggregate analysis (the PCA of Sec. 4, the partials of Sec. 4) each proxy is oriented *within each axis* by the sign of its Spearman correlation with accuracy, falling back to the documented prior direction only when the data correlation is negligible (plain, $\rho{=}-0.01$ on TSS; Sec. 3). This orientation is not bookkeeping: the sign a proxy must be flipped on one axis can differ from the other. The corresponding raw proxy–accuracy relationship is evaluated without orientation in Sec. 5; the oriented residual direction is treated only as sensitivity evidence. The full 29-proxy zoo, its four groups, and the prior directions are listed in Tab. 8.

D. The Kernel Spectral-Bias Map, Restated

We restate, for completeness, the kernel spectral-bias decomposition we use qualitatively in Sec. 3. Following Canatar et al. [4] and Bordelon et al. [3]: for a kernel K with Mercer eigenvalues $\rho_1 \geq \rho_2 \geq \cdots$ and eigenfunctions $\{\phi_i\}$, and a target $f^\star = \sum_i \bar{c}_i \phi_i$, the expected generalization error of kernel ridge regression on n samples takes the modal form

$$\mathcal{E}(n) = \sum_i \bar{c}_i^2 \, \mathcal{R}_i(n; \rho_1, \rho_2, \dots), \tag{1}$$

where each modal factor $\mathcal{R}_i$ depends on the *whole* spectrum and decreases as mode i becomes learnable. The split that matters for us: $\{\rho_i\}$ is label-free (*capacity*); $\{\bar{c}_i^2\}$ requires the labels (*alignment*). *Used in:* Sec. 3 to name the two axes; no quantitative content is imported.

E. Extended Related Work

The body Sec. 2 compresses five themes into one paragraph each; we expand them here, reusing the same references. The unifying gap is that the literature builds proxies that rank but rarely tests which observable factors their scores track or whether those associations are stable across search spaces.

Capacity and expressivity proxies. A first line scores the richness an untrained network can express. NEAR's effective rank [7] and the Dextr spectral condition number [2] (whose cond-sum we reimplement at parity) summarize the at-init feature spectrum, and Zen-NAS [13] scores the response to Gaussian perturbation. These are read as expressivity scores, but none asks how much of every *other* proxy is the same capacity number under another name; Sec. 4 shows a single component aligns strongly with effective rank and carries about half the variance of the zoo.

Trainability and gradient proxies. A second line scores how easily a net will train from one-minibatch gradient, Jacobian, or NTK statistics: ZiCo [12], TE-NAS [6], jacobcov [14], GradSign [18], and MeCo [8]. What is left open is whether, with capacity held fixed, anything remains at init for them to read. Sec. 4 operationalizes this as the second term of the decomposition: a capacity-orthogonal association with early training that is partly predictable from label-free at-init statistics.

Ensembles and joint proxies. A third line mixes capacity and trainability into one recipe (AZ-NAS ensembles four axes [11], InTrain multiplies a capacity term by a gradient-health term [19], and SWAP-NAS scores sample-wise activation patterns [16]) on the implicit assumption that a single combination transports across spaces. Several raw constituents reverse under transport (Sec. 5), while SWAP and AZ-NAS remain positive on both tested axes (Appendix I.6). Recombination therefore requires explicit transport validation rather than a universal sign assumption.

Benchmarks and the cross-space critique. A fourth line records the symptoms without a single account: rankings are inconsistent across spaces and many proxies correlate with accuracy largely through parameter count [15, 17], simple hand-built features rival learned proxies [9], and NAS-Bench-Suite-Zero reports that combining 13 proxies adds up to $+42\%$ over the best single one [10]. Our decomposition organizes these symptoms into testable associations: "proxies $\approx$ params" corresponds to the capacity-associated factor (Sec. 4), part of the remaining complementarity is associated with the early-training target (Sec. 4), and cross-space inconsistency appears as the sign reversals of Sec. 5.

Kernel spectral bias. A fifth line, which we borrow only as a map, reads a label-free statistic's reachable target through the kernel spectral-bias picture of Canatar *et al*. [4] and Bordelon *et al*. [3]: the at-init NTK's kernel-regression error. Our one formal statement (Proposition 1) bounds that *frozen* target, never trained accuracy. Some proxies retain association with early training after the measured capacity controls (Sec. 4). The proposition is motivation; the contribution is the empirical analysis. The body Sec. 2 is the compressed version.

F. Proof of Proposition 1

The body states the bound and defers the derivation here so the two copies cannot drift. The argument is short,

a conditional-expectation projection followed by Cauchy–Schwarz, and the attainment case is recorded so the "resolution limit" reading is exact rather than loose.

Proof. By $0 < \mathrm{Var}(Y) < \infty$ we have $\mathbb{E}[Y^2] < \infty$, so $\mathbb{E}[Y \mid \mathcal{G}]$ exists in L^2 and all covariances below are finite; $0 < \mathrm{Var}(S) < \infty$ justifies the final division. Decompose $Y = \mathbb{E}[Y \mid \mathcal{G}] + \varepsilon$ with $\varepsilon := Y - \mathbb{E}[Y \mid \mathcal{G}]$. By linearity and idempotence of conditional expectation $\mathbb{E}[\varepsilon \mid \mathcal{G}] = 0$, so for any $\mathcal{G}$-measurable S,

$$\mathrm{Cov}(S, \varepsilon) = \mathbb{E}[S\,\varepsilon] - \mathbb{E}[S]\mathbb{E}[\varepsilon] = \mathbb{E}\big[S\,\mathbb{E}[\varepsilon \mid \mathcal{G}]\big] - 0 = 0.$$

Hence $\mathrm{Cov}(S, Y) = \mathrm{Cov}\big(S, \mathbb{E}[Y \mid \mathcal{G}]\big)$, and by Cauchy–Schwarz

$$\mathrm{Cov}(S, Y)^2 = \mathrm{Cov}\big(S, \mathbb{E}[Y \mid \mathcal{G}]\big)^2 \le \mathrm{Var}(S)\ \mathrm{Var}\big(\mathbb{E}[Y \mid \mathcal{G}]\big).$$

Dividing by $\mathrm{Var}(S)\,\mathrm{Var}(Y) > 0$,

$$\mathrm{Corr}^2(S, Y) = \frac{\mathrm{Cov}(S, Y)^2}{\mathrm{Var}(S)\,\mathrm{Var}(Y)} \le \frac{\mathrm{Var}\big(\mathbb{E}[Y \mid \mathcal{G}]\big)}{\mathrm{Var}(Y)} = R^2_{\mathrm{cap}}.$$

That $R^2_{\mathrm{cap}} \in [0, 1]$ is immediate from the law of total variance, $\mathrm{Var}(Y) = \mathrm{Var}\big(\mathbb{E}[Y \mid \mathcal{G}]\big) + \mathbb{E}\big[\mathrm{Var}(Y \mid \mathcal{G})\big] \ge \mathrm{Var}\big(\mathbb{E}[Y \mid \mathcal{G}]\big) \ge 0$. When $R^2_{\mathrm{cap}} > 0$, taking $S = \mathbb{E}[Y \mid \mathcal{G}]$ makes the Cauchy–Schwarz step an equality, and Corr^2 is invariant under affine transformations of S, so the bound is attained; when $R^2_{\mathrm{cap}} = 0$, $\mathbb{E}[Y \mid \mathcal{G}]$ is constant and the bound holds trivially for every admissible S (with $0 < \mathrm{Var}(S) < \infty$). □

Remark (scope). The bound holds for *any* target Y; its content depends entirely on which Y one applies it to. We certify it only for Y = the frozen at-init NTK kernel-regression error. It does *not* bound trained accuracy, and Section 4 shows the tempting "ceiling" reading does not bind on NB201.

G. Extended Results

This appendix collects the per-proxy numbers and sensitivity evidence relocated from Secs. 4 and 5 for length, together with the longer positioning discussion. The results and the headline portability heatmap (Fig. 3) remain in the body; the partial-slot heatmap (Fig. 2, below) and the supporting per-proxy enumerations live here.

G.1. Per-proxy values for the sign-flip

The headline (Sec. 5) is that the *same* raw proxy reverses sign across the two axes. We report the orientation-free accuracy correlation (*accuracy slot*) and, separately, the capacity-residual trainability partial under per-axis outcome-informed orientation. The latter is an orientation-sensitive secondary analysis, not independent evidence for the raw flip.

Accuracy slot. The NB201 columns of Fig. 3 (full numbers in Tab. 3) give $\mathrm{corr}(Q, \mathrm{acc})$ for the proxies whose orientation is not preserved across axes. Our inverse-condition-sum spectral functional moves from $\mathrm{corr}(Q, \mathrm{acc}) = +0.88$ on TSS to -0.77 on SSS; meco from $+0.88$ to -0.79; grasp from $+0.55$ to -0.19 (Fig. 1). These are not weak proxies noisy near zero: cond-sum and meco are among the strongest single proxies on TSS and among the most strongly *anti*-correlated on SSS, so a practitioner who calibrated on topology would, on size, be selecting *against* accuracy.

Trainability slot under outcome-informed orientation. Figure 2 (full numbers in Tab. 3) reports the leakage-clean partial correlation $\mathrm{partial}(Q, T \mid \mathrm{erank}, \mathrm{params})$, the trainability signal a proxy accesses after capacity is regressed out (Sec. 4). jacob-cov, the single best label-free route to trainability on TSS (partial $= +0.604$ $[+0.52, +0.67]$), collapses to -0.124 $[-0.29, +0.06]$ on SSS: its access does not merely weaken, it disappears and turns mildly negative. meco flips from $+0.291$ $[+0.17, +0.40]$ to -0.398 $[-0.51, -0.23]$; cond-sum from $+0.215$ $[+0.09, +0.33]$ to -0.141 $[-0.31, +0.04]$. Our label-free adaptation probe Q-adapt-rot shows the most extreme reversal, from $+0.143$ $[+0.03, +0.24]$ on TSS to -0.602 $[-0.70, -0.47]$ on SSS; gradsign changes from a (standard-arm) $+0.11$ to -0.130 $[-0.29, +0.09]$. Because the orientation itself uses accuracy separately on each axis, these signs are not independent evidence of transport; the raw accuracy-slot reversals remain the orientation-free result.

Narrow survivor under outcome-informed orientation. Applying the conservative two-sided test from Sec. 4 under this rule, the only full-zoo proxy that passes is ntk, in the *trainability partial* slot $\mathrm{partial}(Q, T \mid \mathrm{erank}, \mathrm{params})$ (not the raw-correlation slot): $+0.180$ $[+0.06, +0.30]$ TSS; $+0.241$ $[+0.07, +0.39]$ SSS. At the larger base sample of the standard-proxy arm two more clear the bar, synflow and zico, giving {synflow, zico, ntk}. Under that rule, few topology-axis carriers retain the same sign on the size axis (Fig. 2b). Under the globally fixed rule, both meco and ntk have positive marginal intervals on both axes; this comparison is exploratory and unadjusted, so no orientation-robust uniqueness claim is made.

G.2. Dimension-linked sensitivity evidence

Having shown the raw flip, we test whether representation geometry accompanies it. This account is observational: several scores depend numerically on spectral dimension, and the two axes vary topology and width differently. The controls below support a dimension-linked interpretation but do not identify a mechanism.

	TSS (n=324)			SSS (n=138)		
Proxy	$\rho(Q, \text{acc})$	partial$(Q, T \mid \text{cap})$	$\rho(Q, \text{erank})$	$\rho(Q, \text{acc})$	partial$(Q, T \mid \text{cap})$	$\rho(Q, \text{erank})$
fisher	+0.47	−0.152 [−0.27, −0.04]	+0.513	+0.70	+0.098 [−0.09, +0.25]	+0.728
grad-norm	+0.64	−0.358 [−0.44, −0.26]	+0.747	+0.59	+0.074 [−0.11, +0.24]	+0.648
grasp	+0.55	−0.176 [−0.30, −0.04]	+0.631	−0.19	−0.030 [−0.19, +0.14]	+0.198
snip	+0.64	−0.369 [−0.44, −0.28]	+0.752	+0.86	+0.296 [+0.11, +0.45]	+0.899
synflow	+0.73	−0.126 [−0.22, −0.02]	+0.781	+0.94	+0.332 [+0.14, +0.48]	+0.972
jacob-cov	+0.76	+0.604 [+0.52, +0.67]	+0.726	−0.00	−0.124 [−0.29, +0.06]	+0.025
plain	−0.01	−0.170 [−0.28, −0.06]	−0.013	+0.06	+0.131 [−0.05, +0.29]	+0.069
zico	+0.78	−0.197 [−0.29, −0.09]	+0.880	+0.86	+0.233 [+0.06, +0.40]	+0.887
meco	+0.88	+0.291 [+0.17, +0.40]	+0.856	−0.79	−0.398 [−0.51, −0.23]	+0.910
zen	−0.10	−0.049 [−0.15, +0.05]	+0.146	+0.90	+0.517 [+0.37, +0.63]	+0.932
ntk	−0.78	+0.180 [+0.06, +0.30]	+0.774	−0.20	+0.241 [+0.07, +0.39]	+0.139
gradsign	+0.78	−0.135 [−0.23, −0.03]	+0.810	+0.86	−0.130 [−0.29, +0.09]	+0.937
cond-sum	+0.88	+0.215 [+0.09, +0.33]	+0.893	−0.77	−0.141 [−0.31, +0.04]	+0.830
stable-sum	+0.86	+0.109 [−0.02, +0.23]	+0.971	+0.68	−0.014 [−0.17, +0.15]	+0.740
$R_{\alpha=0.5}$	+0.84	−0.266 [−0.36, −0.13]	+0.993	+0.90	−0.295 [−0.44, −0.03]	+0.997
$R_{\alpha=2}$	+0.88	+0.256 [+0.12, +0.37]	+0.995	+0.91	+0.067 [−0.13, +0.26]	+0.995
$R_{\alpha=4}$	+0.88	+0.245 [+0.12, +0.35]	+0.989	+0.89	−0.010 [−0.19, +0.17]	+0.976
$R_{\alpha=8}$	+0.88	+0.223 [+0.09, +0.34]	+0.985	+0.88	+0.006 [−0.14, +0.18]	+0.959
$R_{\alpha=16}$	+0.87	+0.205 [+0.08, +0.31]	+0.984	+0.87	+0.015 [−0.15, +0.19]	+0.946
Q-ntk-logcond	+0.30	−0.121 [−0.24, +0.01]	+0.405	−0.53	−0.475 [−0.62, −0.30]	+0.659
Q-ntk-logtrace	−0.36	−0.075 [−0.18, +0.03]	+0.404	+0.29	−0.032 [−0.21, +0.14]	+0.292
Q-ntk-effrank	−0.50	+0.105 [+0.00, +0.21]	+0.565	+0.58	−0.343 [−0.52, −0.14]	+0.667
Q-ntk-loglmax	−0.36	−0.078 [−0.18, +0.04]	+0.406	−0.11	−0.198 [−0.32, −0.06]	+0.196
Q-zico-rand	+0.31	+0.207 [+0.08, +0.32]	+0.310	+0.09	+0.121 [−0.06, +0.29]	+0.120
Q-zico-rot	+0.40	+0.183 [+0.06, +0.30]	+0.392	+0.07	−0.045 [−0.20, +0.14]	+0.077
Q-coh-rand	−0.48	−0.072 [−0.16, +0.03]	+0.535	−0.09	+0.075 [−0.13, +0.26]	+0.133
Q-coh-rot	+0.08	+0.128 [+0.04, +0.21]	+0.038	−0.24	−0.205 [−0.38, −0.01]	+0.282
Q-adapt-rot	+0.10	+0.143 [+0.03, +0.24]	+0.041	+0.65	−0.602 [−0.70, −0.47]	+0.809
Q-zico-oracle^{L}	+0.35	+0.253 [+0.13, +0.37]	+0.334	+0.05	+0.040 [−0.13, +0.22]	+0.060

Table 3. **Complete per-proxy NB201 results (full-zoo)** on the topology (TSS) and size (SSS) axes; the partial columns are rendered as body Fig. 2. $\rho(Q, \text{acc})$ is shown raw and ^{L} Q-zico-oracle uses labels (excluded from all label-free claims).

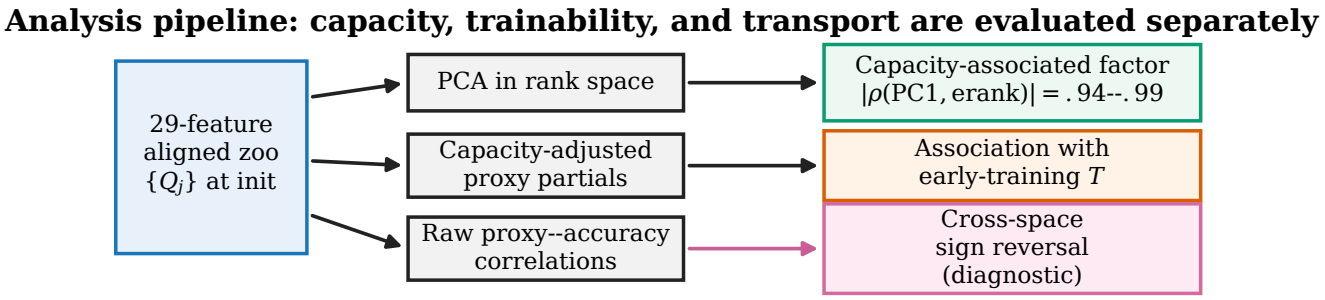


Figure 1. **Analysis pipeline.** A 29-feature aligned zoo is summarized by a capacity-associated factor (PC1 scores track effective rank) and a capacity-orthogonal trainability residual. Raw proxy/accuracy reversals are orientation-free; per-proxy residual directions use the stated orientation rule and are treated as sensitivity evidence. Proposition 1 concerns only the frozen at-init NTK target Y.

Observed axis dependence. Several reversing scores are functionals of the representation spectrum (condition number, feature covariance, or Jacobian/NTK quantities) whose numerical value depends on representation geometry. TSS holds width fixed while varying topology; SSS varies width directly. This makes dimension-linked score behavior plausible, but the observational comparison cannot determine which architectural quantity causes the reversal.

TSS (PC1, 52.9% var)		**SSS** (PC1, 49.4% var)	
Proxy	loading	Proxy	loading
$R_{\alpha=16}$	+0.246	$R_{\alpha=0.5}$	+0.260
$R_{\alpha=8}$	+0.246	$R_{\alpha=2}$	+0.260
$R_{\alpha=4}$	+0.246	synflow	+0.257
stable-sum	+0.245	$R_{\alpha=4}$	+0.256
$R_{\alpha=2}$	+0.244	$R_{\alpha=8}$	+0.253
$R_{\alpha=0.5}$	+0.233	$R_{\alpha=16}$	+0.250
cond-sum	+0.231	gradsign	+0.247
snip	+0.225	meco	+0.245
grad-norm	+0.225	zen	+0.244
ntk	+0.224	snip	+0.240
meco	+0.215	zico	+0.235
zico	+0.211	Q-adapt-rot	+0.227
gradsign	+0.201	cond-sum	+0.225
jacob-cov	+0.194	stable-sum	+0.200

Table 4. **Leading principal component of the oriented zoo (top-14 loadings;** n=317/138 **complete cases),** which loads broadly across the spectral and gradient proxies; corr(PC1, erank)=+0.937 (TSS) / +0.986 (SSS).

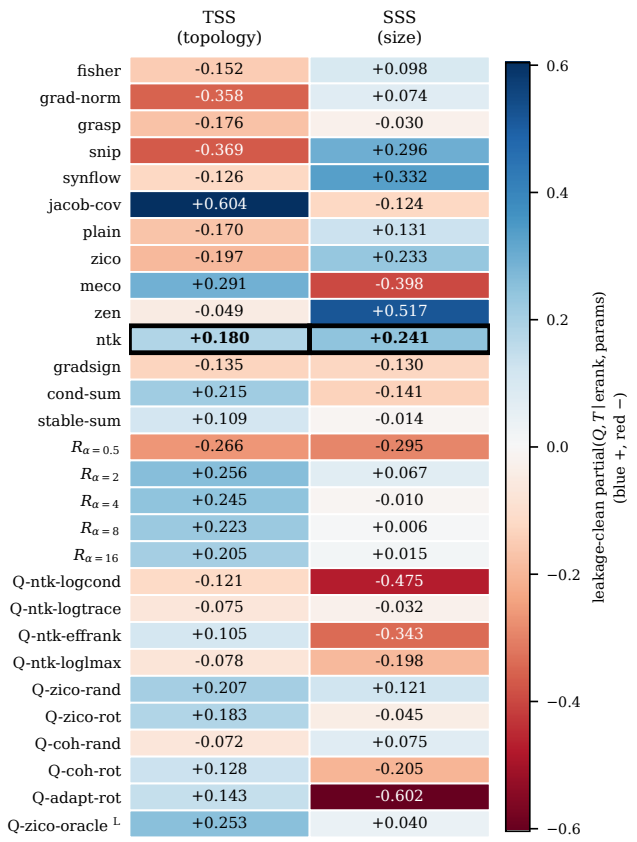


Figure 2. **Outcome-oriented trainability partials.** Leakage-clean partial partial$(Q, T \mid \text{erank}, \text{params})$ for every proxy on TSS (x) versus SSS (y), after outcome-informed per-axis orientation. The global orientation control shows that directional quadrant membership is sensitive to this rule (Appendix I.1).

Scope of the SSS-side association. On the SSS (size/width) axis, effective-rank conditioning greatly weakens the observed association, suggesting a dimension/capacity-linked component rather than an independent mechanism. This is not a mediation test or a validated representation-dimension match: the available cached count is not a validated dimension field, and its support does not overlap across axes (Appendix I.3). The cond-sum SSS raw correlation corr(Q, acc) is -0.768; it collapses when we condition on effective rank (-0.047, 95%CI $[-0.242, +0.128]$, spanning zero) and on params+erank jointly (-0.048, $[-0.260, +0.130]$), while conditioning on *params alone* leaves it intact (-0.317, $[-0.478, -0.128]$). The meco/jacob-cov/grasp SSS partials collapse under the same effective-rank control. The TSS-side positive partials are *not* explained away by capacity: jacob-cov $+0.604$ and cond-sum $+0.215$ both survive BH-FDR and Holm correction.

The spectral family exposes the dependence. Within our own spectral family the descriptive dependence is visible (Sec. 3): the inverse-condition-sum cond-sum, a pure conditioning functional, is exactly the one that flips ($+0.887$ TSS / -0.781 SSS on the n=500 spectral-arm population; the full-zoo n=324 values used in the body and conclusion are the rounded $+0.88$ / -0.77), whereas the Rényi effective-rank functionals R_α, which integrate over the whole spectrum rather than its conditioning, remain positive on both axes (e.g. $R_{\alpha=1}$, NEAR's effective rank, $+0.864$ / $+0.911$; $R_{\alpha=2}$, $+0.879$ / $+0.912$). The choice of spectral summary is associated with whether a proxy ports across axes. Frozen cross-space transfer also degrades asymmetrically: a linear model fit on TSS assigns cond-sum a positive weight and reaches transfer Spearman $+0.5405$ on SSS, whereas a model fit on SSS down-weights the sign-unstable functional and transfers back to TSS at $+0.8629$.

G.3. Generalization across search spaces (per-space values)

Sign reversals are also observed outside the two NB201 axes (Sec. 5). This subsection lists the four strictly supported instances, then reports multiplicity controls for the larger grid.

Under the bootstrap-CI quarantine (2000 resamples; CI-separated = the two spaces' 95% intervals disjoint), four flips are load-bearing: jacob-cov $+0.78\,[+0.69, +0.85]$ on TransNAS-scene vs. $-0.53\,[-0.67, -0.35]$ on NB101; grasp $+0.56\,[+0.51, +0.60]$ on NB201-CIFAR-100 vs. $-0.57\,[-0.69, -0.43]$ on TransNAS-scene; zen $+0.67\,[+0.55, +0.76]$ on TransNAS-scene vs. $-0.09\,[-0.16, -0.01]$ on NB201-CIFAR-100; and

Proxy	NB101	NB201-c10	NB201-c100	NB201-IN16	NB301	TNAS-obj	TNAS-sc
grad-norm	+0.509	+0.595	+0.640	+0.580	**−0.153**	+0.389	+0.650
snip	+0.539	+0.595	+0.642	+0.579	**−0.162**	+0.431	+0.684
fisher	+0.501	+0.394	+0.427	+0.361	**−0.338**	+0.464	+0.655
jacob-cov	**−0.526**	+0.721	+0.700	+0.677	**−0.075**	+0.604	+0.783
synflow	+0.693	+0.737	+0.766	+0.757	+0.221	+0.466	+0.647
nwot	+0.323	+0.755	+0.800	+0.775	+0.565	+0.320	+0.517
zen	+0.651	**−0.065**	**−0.086**	**−0.017**	+0.470	+0.495	+0.669
grasp	**−0.102**	+0.510	+0.556	+0.504	+0.205	**−0.390**	**−0.566**
l2-norm	+0.547	+0.678	+0.719	+0.703	+0.461	+0.268	+0.495
plain	+0.047	**−0.082**	**−0.081**	**−0.053**	**−0.295**	+0.516	+0.326
params	+0.670	+0.717	+0.735	+0.702	+0.502	+0.387	+0.577
flops	+0.478	+0.717	+0.735	+0.702	+0.470	**−0.048**	**−0.125**
epe-nas	**−0.080**	+0.704	+0.691	+0.665	**−0.011**	+0.520	+0.432

Table 5. **Cross-benchmark portability matrix:** Spearman corr(Q, acc) point estimates for each standard proxy (**bold**=negative), rendered as body Fig. 3. Sample sizes are NB101 100, NB301 50, and TransNAS 150. NB201 is 1000 except jacob-cov, zen, and epe-nas (977) and nwot (984). Strict-case intervals are in Appendix G.3.

Dataset	Features	R^2_{in}	R^2_{cv}
CIFAR-10	params	0.526	+0.526
CIFAR-10	params+T	0.779	+0.775
CIFAR-10	params+erank	0.732	+0.730
CIFAR-10	params+T+erank	0.831	+0.828
CIFAR-100	params	0.541	+0.541
CIFAR-100	params+T	0.748	+0.743
CIFAR-100	params+erank	0.780	+0.778
CIFAR-100	params+T+erank	0.843	+0.840
IN16-120	params	0.480	+0.482
IN16-120	params+T	0.776	+0.777
IN16-120	params+erank	0.726	+0.727
IN16-120	params+T+erank	0.837	+0.837

Table 6. **The capacity/trainability decomposition replicates across the three NB201 datasets** (n=324; rank-R^2 of accuracy, in-sample and leakage-clean 5-fold CV); the R^2_{cv} slice is body Tab. 1. This arm uses curve-derived accuracy and the train-accuracy T (Appendix H.5).

Population	TSS	SSS
Full-zoo table intersection (before finite mask)	324	138
PCA-complete 29-feature population	317	138
Standard-proxy base table	1000	425
of which erank / T present	324	138

Table 7. Sample sizes per search-space axis. The full-zoo join precedes analysis-specific finite-value masks; the fixed PCA mask leaves 317/138 complete cases. The standard-proxy population covers more architectures but fewer proxies. The SSS arm (n=138) is the binding sample-size constraint.

Group	#	Members
Standard	12	fisher, grad-norm, grasp, snip, synflow, jacob-cov, plain, zico, meco, zen, ntk, gradsign
Spectral	7	cond-sum, stable-sum, $R_{\alpha=0.5,2,4,8,16}$
Label-free probes	9	Q-ntk-{logcond, logtrace, effrank, loglmax}, Q-zico-{rand, rot}, Q-coh-{rand, rot}, Q-adapt-rot
Label oracle (flagged)	1	Q-zico-oracle
Total	**29**	

Table 8. The 29-feature aligned zoo. $R_{\alpha=1}$ is held out as the effective-rank capacity anchor. Per-axis outcome orientation is accompanied by raw and globally fixed controls (Sec. 3). The single label-using oracle is held apart as an upper reference.

plain $+0.33\,[+0.15, +0.47]$ on TransNAS-scene vs. $-0.08\,[-0.14, -0.02]$ on NB201-CIFAR-10. For jacob-cov and grasp both sides are large; for zen and plain the negative side is mild but its CI excludes zero (large-n NB201 arms). With all 13 proxies scored on the TransNAS-object column (n=150), these four cases remain unchanged: flops and epe-nas reverse only to near-zero on TransNAS (CIs spanning zero) and stay suggestive, and only NB301 (n=50, disjoint 0/50) remains excluded as degenerate. The full 13×7 portability matrix with bootstrap intervals is in Appendix H.5.

Multiplicity control. The four strict instances remain supported under the reported multiplicity controls. On the per-proxy early-training partial grid (58 cells, partial$(Q, T_{\text{slope}} \mid \text{erank}, \text{params})$ over 29 proxies $\times$ two

Proxy	Space	ρ	perm p	n
jacob-cov	NB101-CIFAR-10	-0.526	2.0×10^{-4}	100
grasp	TransNAS-scene	-0.566	2.0×10^{-4}	150
zen	NB201-CIFAR-100	-0.086	6.2×10^{-3}	977
plain	TransNAS-scene	$+0.326$	2.0×10^{-4}	150

Table 9. **Each load-bearing flip is FDR-significant** (the flip cell shown; all four pass BH-FDR, three also Holm).

Benchmark / dataset	n	rank-R^2(acc)		Δ_T	res. gain
		cap	$+T$	share	(cap-surv.)
NB201 / CIFAR-10	1000	0.51	0.79	$+0.28$	$+33\%$
NB201 / CIFAR-100	1000	0.54	0.75	$+0.21$	$+44\%$
NB201 / IN16-120	1000	0.49	0.78	$+0.29$	$+49\%$

Table 10. **The descriptive capacity-plus-T pattern recurs** across NB201's three datasets (params-anchored, leakage-clean): the capacity baseline explains 0.49–0.54 of accuracy rank and ground-truth T adds a positive increment Δ_T.

spaces, 5000 Freedman–Lane permutations per cell), 34 cells survive BH-FDR at α=0.05 (25 survive the conservative Benjamini–Yekutieli, 23 survive Holm–Bonferroni). On the portability grid (91 finite cells, including the full TransNAS-object column), 78 survive BH-FDR and 72 survive Holm. Each of the four load-bearing flips is itself FDR-significant (Tab. 9); the apparent grad-norm/snip/fisher/epe-nas candidate flips occur only in small-n=50 NB301-CIFAR-10 attenuation cells ($|\rho| \leq 0.34$), and flops reverses only to near-zero on TransNAS; none meets the strict support rule. The ntk T_{slope} partial survives BH-FDR in *both* spaces (TSS $+0.180$, perm p=0.0016; SSS $+0.241$, perm p=0.0054).

The capacity/trainability decomposition itself replicates across the three NB201 datasets (Appendix H.5, Tab. 6): parameters explain $R^2 \approx 0.48$–0.54, and adding either the trainability target T ($\sim +0.25$) or effective rank ($\sim +0.23$) lifts R^2 comparably; the joint model including both reaches ~ 0.83 (these are nested-model R^2 values, not additive terms of Eq. (1)).

G.4. Replication across the NB201 datasets

We apply the same analysis protocol (Sec. 4) to NB201's three datasets. Table 10 shows the descriptive pattern recurs: capacity is the dominant baseline, adding ground-truth T increases rank-R^2, and a capacity-surviving complementarity gain persists. The capacity-anchored trainability partial reproduces the headline value: with effective rank as the capacity control (n=324), partial$(T, \text{acc} \mid \text{erank}, \text{params})$=$+0.608/+0.536/+0.635$ on CIFAR-10/CIFAR-100/ImageNet16-120 (cf. $+0.602$ in the main analysis), and the spectral PCA again aligns with effective rank (corr(PC1, erank)=$+0.96$, PC1 $\sim 69\%$ variance).

G.5. Empirically instantiating Proposition 1: the NTK target

The Proposition is a yardstick, not a headline, and this subsection earns that status by measurement rather than assumption. We compute its target, show what a label-free dictionary explains, report a negative result (even the frozen kernel is not a pure-capacity object), and test whether this at-init target is a useful surrogate for trained accuracy. It is not: the bound prices Y exactly and remains silent on accuracy, as scoped in the body.

Computing the target. Proposition 1's target Y is usually only *described*; here we *compute* it (at init). For each of the 324 full-zoo NB201-TSS architectures we form the weight-only empirical NTK Gram over a fixed CIFAR-10 batch (m=256 samples with $m<P$, so the Gram is rank-$\leq m$ and KRR interpolates; 5 batches) and take Y as the held-out kernel-ridge-regression MSE against the real labels (closed-form; the quantity Eq. (1) describes). This is a weight-only at-init NTK at one operating point, not the full NTK. Y is rank-stable across batches (Spearman$(\text{batch}_0, \text{batch}_4)$=0.85).

What a finite label-free dictionary explains. On the across-architecture ensemble the full σ-algebra contains the architecture identity, so Y is arch-determined and the proposition's R^2_{cap}=1 exactly, vacuous (Sec. 3, point (iii)). The non-trivial object is therefore the variance of Y explained by a *pre-declared finite label-free dictionary* F (30 features: NTK-Gram spectral summaries, our spectral proxies, and the gradient/zoo proxies): $R^2(F \to Y)$=0.81 (rank-CV R^2=0.8112; the matching L^2 in-sample Pearson estimate is $R^2 \approx 0.6645$). We do *not* call this $R^2(F \to Y)$ a "resolution limit": it is one dictionary's explained variance, not the proposition's R^2_{cap}. On these data every single proxy's squared rank correlation with Y stays below the dictionary's rank-CV R^2 (max single 0.68 vs. 0.81); no proposition forces this: it is an empirical observation, not a Pearson instance of Proposition 1.

Even the frozen kernel is not a pure-capacity object. Among the proxies (excluding functions of K_a), the *gradient* family predicts Y better than the *spectral/capacity* family (best 0.68 vs. 0.42; family gap -0.2591 $[-0.3367, -0.1862]$, CI-separated). The naive "Y is governed by the spectrum" reading is thus not supported even for the frozen-kernel target. We report this as a negative result.

Coherence: the at-init NTK target is a weak surrogate for trained accuracy. The same dictionary explains Y (0.81) and trained accuracy (rank-CV R^2=0.89) compa-

PC1 of the oriented zoo	TSS	SSS
Var(PC1) (full zoo)	52.9%	49.4%
Var(PC1+PC2)	66.8%	56.9%
Var(PC1) (raw-z)	49.6%	49.8%
Var(PC1) (no-Rényi, 12 proxies)	56.0%	55.1%
corr(PC1, erank) (no-Rényi)	+0.881	+0.968

Table 11. **One factor ≈ effective rank carries half the zoo:** PC1 variance share and its alignment with effective rank, robust to a Rényi-free 12-proxy control (Sec. 4).

rably, yet the per-architecture at-init NTK error Y correlates only ρ=0.40 [0.29, 0.49] with trained accuracy (robust across the ridge: ρ=0.11/0.40/0.47 for λ_K=0.01/0.1/1). These NB201-TSS (CIFAR-10) observations do not support treating this frozen at-init NTK target as a quantitative surrogate for trained accuracy: Proposition 1 prices Y exactly but is silent on accuracy, consistent with the narrow scope in Sec. 3 and Sec. 4. The best single proxy reaches R^2=0.78 against accuracy, *not* bounded by $R^2(F \to Y)$=0.81: different targets, no proposition forcing the at-init ceiling onto accuracy.

G.6. PC1 correlations

Section 4 reads the dominant axis of the zoo as capacity. The numbers behind that reading use the fixed 29-feature finite mask (n=317/138): the leading PCA component of the oriented zoo is strongly associated with effective rank, accuracy, and parameter count: corr(PC1, erank)=+0.937 (TSS) / +0.986 (SSS); corr(PC1, acc)=+0.848 / +0.903; corr(PC1, params)=+0.817 / +0.932 (Fig. 2a). The variance shares and the Rényi-free control are in Tab. 11; the full loadings are Tab. 4.

G.7. Per-proxy trainability partials

Section 4 reports capacity-orthogonal associations between several at-init scores and the early-training target. The per-proxy partials identify *which* statistics carry the signal: on TSS, partial$(Q, T \mid \text{erank}, \text{params})$ is +0.604 for jacob_cov ([+0.52, +0.67]), +0.291 for meco, +0.180 for ntk, and +0.105 for the NTK effective rank Q_ntk_effrank; on SSS synflow reaches +0.332, zen +0.517, and zico +0.233 (zico is itself negative on TSS, −0.197, an instance of the cross-space sign-flip of Sec. 5). The label-using oracle Q_zico_oracle (+0.253 TSS) is flagged and excluded from all label-free claims. A handful of standard proxies (the Jacobian/NTK family and meco on topology, gradient proxies on size) thus retain association with early training after the measured controls, while most do not.

Relation to NAS-Bench-Suite-Zero. Krishnakumar et al. [10] report that an ensemble of zero-cost proxies improves over the best single proxy by roughly +42%, attributing it to complementarity without isolating the associated factors. In our corresponding NB201 analysis, the all-proxy ridge lifts CV-R^2 over the best single proxy by +0.1074 (+13.8%) on TSS and +0.0091 (+1.0%) on SSS[1]. Section 4 shows that, after the measured capacity controls are removed, the surviving gain remains associated with T. On TSS this is consistent with a partly predictable early-training residual; on SSS the evidence is weaker (n=138, a +1.0% pre-residual replicate and +0.064 surviving gain). Per-proxy residual directions are sensitive to the orientation rule, while the raw transport failures are treated in Sec. 5.

G.8. Field positioning (extended)

The cross-axis sign-flip is a within-benchmark instance of the inconsistency repeatedly reported for zero-cost proxies *across* benchmarks: there is no universally best proxy, rankings vary by benchmark, and many correlate strongly with parameter count [9, 10, 15, 17]. Our decomposition organizes those observations into a capacity-associated factor (Sec. 4) plus a capacity-orthogonal early-trainability association (Sec. 4). Several raw proxies change sign with the axis, but the recent controls show that this is not universal (Sec. 5 and Appendix I.6). Methods that recombine proxies [11] therefore need transport validation; portable init-time-trainability claims [19] remain outside the methods we could reproducibly evaluate.

H. Experimental Details

This section records the populations, the spectral-arm parity check, the leakage-clean cross-validation, the bootstrap protocol, and the cross-space join caveats.

H.1. The proxy zoo and populations

The full-zoo join is the intersection of all four measurement tables (spectral, comparison, premise, probe), yielding n=324 TSS and n=138 SSS rows before analysis-specific finite-value masks. The fixed 29-feature PCA mask leaves n=317/138 complete cases. The standard-proxy population uses only the 13 standard metrics, covering more architectures (n=1000 TSS, 425 SSS), with erank and T present on the full-zoo subset within each. Three proxies are computed on separate single-architecture infer.tiny nets and merged in; two return a degenerate value on chance-level (test accuracy ≤10%) architectures, nwot's ReLU activation kernel is singular and epe_nas's per-class Jacobian is constant, so those rows are dropped (nwot n=984, epe_nas n=977), while the parameter-only l2_norm reaches the

[1] Our gain is a lower bound: the best-single baseline is selected in-sample by arg max CV-R^2, which inflates the baseline and shrinks the reported gain.

Spectral functional	TSS	SSS
cond-sum (inv. cond., Dextr)	+0.887	**−0.781**
$R_{\alpha=1}$ (effective rank, NEAR)	+0.864	+0.911
$R_{\alpha=2}$	+0.879	+0.912
$R_{\alpha=4}$	+0.879	+0.901
$R_{\alpha=16}$	+0.874	+0.874

Table 12. **Within the spectral family, only the conditioning functional flips** (corr(Q, acc), n=500 per axis): cond-sum reverses sign (**bold**) while the whole-spectrum Rényi effective ranks stay sign-stable.

full n=1000. Dropping the dead architectures is conservative (imputing them at the proxy minimum would only *raise* the correlation). Proxies are sign-oriented per axis (Sec. 3); complete per-proxy partials are Tab. 3, PC1 loadings Tab. 4.

H.2. Faithfulness of the spectral arm

Our cond-sum reproduces the official Dextr inverse-condition-number statistic; the feature-extraction hooks raised 0 exceptions across all architectures. The sourced parity check is on the spectral arm: on TSS, corr(cond-sum, acc)=+0.8868 (+0.8805 on live architectures with acc>15%), consistent with the published Dextr value $\approx$ +0.88. This TSS parity makes a simple extraction mismatch less plausible, but it does not validate a causal explanation for the SSS result. The Rényi sweep R_α, $\alpha \in \{0.5, 1, 2, 4, 8, 16\}$, has an official reference only at α=1 (NEAR's effective rank); $\alpha \neq 1$ is treated as exploratory. The spectral-functional sign-stability summary is Tab. 12.

H.3. Decomposition methodology (leakage-clean)

For the accessibility result (Sec. 4) we regress capacity {erank, params} out of *both* accuracy and every proxy *inside each cross-validation train fold*: ranks are computed within the train fold, test rows are mapped by `searchsorted`, and ridge fit plus standardization use train folds only. The held-out residual prediction therefore never touches the held-out target. Reported R^2 values are out-of-fold. The all-proxy-over-best-single complementarity gain is a lower bound: the best-single baseline is selected in-sample by arg max CV-R^2, which inflates it. "Ceiling" lifts subtract a best-of-K selection noise floor (p_{95} in Sec. 4). The trainability residual (Sec. 4) is also robust to how T is defined: the capacity-controlled partial partial$(T, \text{acc} \mid \text{erank}, \text{params})$ stays positive on both axes under either alternative: loss-based and slope-based T give +0.610 / +0.645 (TSS) and +0.637 / +0.506 (SSS).

H.4. Bootstrap intervals.

For the NB201 decomposition and portability analyses, confidence intervals use nonparametric paired resampling over architectures with fixed seeds: the decomposition uses 1000 resamples and the strict portability screen (Appendix G.3) uses 2000, with percentile endpoints. The validity controls in Appendix I state their own units and draw counts.

H.5. Cross-space joins and caveats

The cross-benchmark arm (Sec. 5, Tab. 5) joins proxy values with final benchmark accuracy for the raw portability map. The deep decomposition is run only where the exact fields of Eq. (1) coexist. NB301's loss slope is retained only as an explicitly non-equivalent auxiliary field: it is not substituted for the train-accuracy target T, and NB301 is ineligible for the external decomposition audit. Key methodological points and caveats follow:

- **NB201 index alignment** between the proxy tables and the training-curve pickles was verified by accuracy agreement (mapping base index to the curve pickle's iteration order): final accuracies agree to mean absolute difference 0.40pp, with 95% of 1000 architectures within 1pp.
- **NB201 nwot/l2-norm/epe-nas** are computed at initialization on the single-architecture NB201 networks (l2-norm needs the single-architecture net, as the supernet gives a constant parameter count). Their sample sizes are nwot n=984, l2-norm n=1000, epe-nas n=977; the acc=10% dead architectures are listwise-dropped (conservative).
- **NB201 cross-dataset columns** (CIFAR-100, ImageNet16-120) reuse the CIFAR-10-computed proxy values against the other dataset's accuracy (dataset-agnostic on TSS), measuring cross-dataset portability.
- **NB301** has n=50 proxy architectures, *disjoint* from the 11221-architecture curve set (genotype matches 0/50); we use the proxy json's own accuracy and report its small-n flips as suggestive.
- **TransNAS-Bench-101** is joined per task by record index; all 13 standard proxies are scored on both the *object* and *scene* tasks (n=150 each), while the *jigsaw* task is omitted (its stem lacks the `.act` hooks fisher requires). Our spectral proxies (R_α, cond-sum, …) exist for NB201 only and are N/A elsewhere.

The decomposition replication across the three NB201 datasets is Tab. 6 (curve-derived accuracy and the uniform five-point T, so its values differ slightly from Sec. 4 but preserve the ordering).

I. Validity and Robustness Controls

This section reports orientation, sampling, dimension, search, external-coverage, and recent-proxy controls. Intervals retain their experiment-specific resampling unit; algorithmic subsamples/resamples are not treated as independent experiments. No result below depends on an external

PCA input/orientation	variance T/S	$\lvert\rho(\text{PC1}, \text{erank})\rvert$ T/S	loading cosine
rank, per-axis outcome	.529/.494	.937/.986	.869
rank, raw/global	.529/.494	.937/.986	.461
raw-z, per-axis outcome	.496/.498	.936/.981	.875
raw-z, raw/global	.496/.498	.936/.981	.445

Table 13. **Orientation affects cross-axis loading direction, not factor strength.** T/S denote the NB201 topology/size axes.

link; the evidence is documented directly here.

I.1. Orientation control

The per-axis outcome-informed analysis orients every proxy separately on each axis using its accuracy correlation. Multiplying columns by fixed signs is an orthogonal reflection: it cannot change PCA eigenvalues and changes scores only up to the arbitrary component sign. Raw and globally fixed analyses confirm this invariance but show that directional loading agreement was overstated (Tab. 13). We therefore retain the dominant capacity-associated factor, not an unchanged PC1 or loading direction. The globally oriented trainability partials for meco ($+0.291$ $[0.177, 0.397]$ / $+0.398$ $[0.233, 0.513]$) and ntk ($+0.180$ $[0.056, 0.295]$ / $+0.241$ $[0.077, 0.388]$) are exploratory marginal intervals without multiple-testing certification. These marginal intervals use 2,000 percentile-bootstrap draws per axis.

I.2. Sampling, missingness, and matched sample count

The standard-proxy populations contain 1,000 TSS and 425 SSS architectures. The spectral/premise/probe cohorts each contain 500 per axis; complete-case joins with the base populations yield 324 TSS and 138 SSS records. The join loss is cohort overlap, not row-wise disagreement among already joined tables. The upstream 500-cohort sampling algorithm cannot be recovered from the cached artifacts.

The SSS intersection is mildly shifted toward higher accuracy and larger models: the mean accuracy difference is $+0.285$ percentage points (raw $p = .032$, Holm $p = .096$), and parameter-related differences likewise do not pass the within-axis Holm correction. We therefore retain a representativeness limitation rather than conclude that selection is absent.

After first applying the fixed 29-feature finite mask (317 eligible TSS records), we drew 1,000 exact without-replacement samples of $n = 138$. The full summary is in Tab. 14. All three headline raw reversals retain the TSS-positive/SSS-negative direction in 1,000/1,000 samples. These are algorithmic subsampling summaries, not 1,000 independent trials.

I.3. Cached-length and effective-rank controls

On balanced $n = 500/500$ cohorts, cached normalizations retain the full-population cond-sum sign difference

Matched-n quantity	95% subsampling interval
PC1 variance	$[.497, .559]$
$\rho(\text{PC1}, \text{erank})$	$[.918, .953]$
outcome-oriented cosine vs. SSS	$[.835, .889]$
capacity rank-R^2	$[.666, .798]$
capacity+T rank-R^2	$[.785, .878]$
T increment	$[.060, .138]$
cond-sum/meco/grasp direction	$1000/1000$ each

Table 14. **Matched TSS sample size does not explain the headline results.** The fixed SSS comparison remains $n = 138$.

Score correlation	TSS	SSS
raw cond-sum	$+.887\,[+.860, +.906]$	$-.781\,[-.815, -.743]$
cond-sum/cached length	$+.842\,[+.810, +.866]$	$-.785\,[-.820, -.746]$
cond-sum/effective rank	$+.418\,[+.340, +.491]$	$-.885\,[-.902, -.861]$
effective-rank common-range sensitivity	$-.318\,[-.500, -.126]$	$-.885\,[-.902, -.861]$

Table 15. **Dimension-linked sensitivities.** The final row is an unmatched common-range sensitivity, not a causal or exact dimension match.

(Tab. 15). Within SSS, however, the partial of the raw score is -0.079 after effective-rank control and -0.068 after joint cached-length, effective-rank, and parameter controls. Cached-length supports are disjoint (TSS 142–502; SSS 118–122), so exact cross-axis matching is not identifiable. An unmatched effective-rank common-range sensitivity uses only 103 TSS records and gives cond-sum/effective-rank correlations $-0.318/-0.885$, removing the cross-axis sign difference. Because the cached count is not a validated representation dimension, an exact dimension-matched analysis cannot be performed from the available artifacts. Within exact cached-length strata, all 10 TSS strata with $n \geq 20$ remain positive ($\rho = +.396$ to $+.843$), while the two large SSS strata remain negative (length 120: $-.849$; length 122: $-.766$). All marginal intervals in Tab. 15 use 2,000 bootstrap draws per estimand.

I.4. Search consequence and sensitivity protocols

The screen uses target-CALIB cached accuracy to reject a source proxy whose sign does not transport, and then evaluates a disjoint target-EVAL set. It is not fully label-free. Table 16 gives the source-best comparisons for all K values. The pre-specified $K = 10$ primary point is a moderate budget among $K = 1/5/10/50$; retaining all four avoids favorable-budget selection. At $K = 10$, percentile intervals under the target-position resampling protocol (1,000 algorithmic draws) are $[4.386, 6.435]$ and $[0.626, 3.133]$ for the two traps; the two null intervals are $[0, 0]$. The identity-disjoint sensitivity first partitions canonical identities and has zero overlap; its trap intervals are $[5.057, 5.599]$ and $[1.487, 2.208]$, with both nulls again $[0, 0]$. The protocols use different resampling units and are not pooled.

Screen minus source-best (pts)	K=1	K=5	K=10	K=50
TSS→SSS trap	+7.826	+6.622	+5.402	+3.001
SSS→TSS null	0	0	0	0
scene→NB101 trap	+1.108	+1.554	+1.973	+1.563
scene→NB201 null	0	0	0	0

Table 16. **Two named traps and two null directions.** Values are point costs under the frozen target-CALIB/target-EVAL protocol, not average gains.

Population	joined rows	exact T/final acc.	effective rank
NB101	9,781	0	0
NB301	11,221	0	0
TransNAS accuracy tasks (4)	$4 \times 4,096$	$4 \times 4,096$	0
TransNAS task-native metrics (3)	$3 \times 4,096$	0	0

Table 17. **External field-coverage audit.** Joined rows do not imply eligibility for the full decomposition.

At K=10 under the target-position resampling protocol, screen-minus-baseline point differences for the TSS→SSS trap are -0.009 (params), -0.054 (SynFlow), -0.044 (fixed rank ensemble), and $+1.989$ (random). For scene→NB101 they are $+0.080$, 0, $+1.127$, and $+1.448$. Thus the screen addresses the named source-best traps but does not dominate every baseline. Percentile intervals are reported without inferential p-values; resampling sign mass is only an algorithmic diagnostic.

I.5. External decomposition coverage audit

The external audit joins 49,674 canonical records (Tab. 17). Official inputs contain the 13-field common proxy set, but effective-rank coverage is zero. NB101/NB301 also lack exact early-training accuracy T. Four TransNAS accuracy tasks have exact T and final accuracy; room layout, normal, and autoencoder instead expose task-native early/final metrics. No population jointly contains all four field classes—sufficient proxy coverage, effective rank, exact early T, and exact final accuracy. Eq. (1) therefore has zero eligible external rows, and no surrogate decomposition is reported.

I.6. Recent-proxy controls

SWAP [16] and AZ-NAS [11] were evaluated on the common 1,425-architecture TSS/SSS population. Table 18 uses 10,000 architecture-identity bootstrap draws. SWAP is one sign-stable recent control and does not represent every recent method. L-SWAG [5] and InTrain [19] remain unevaluated because no official, licensed, auditable implementation was located as of August 25, 2026; no approximate substitute was used.

J. Reproducibility

Every body number traces to a logged source. Stochastic procedures are deterministic under their recorded, protocol-specific seed schedules (rather than one global seed) and use the public NAS-Bench-201 / NATS-Bench, NAS-Bench-101, NAS-Bench-301, and TransNAS-Bench-101 benchmarks. We will release the analysis code and per-architecture values.

Proxy	TSS ρ [95%]	SSS ρ [95%]
SWAP	.798 [.766, .825]	.778 [.731, .819]
AZ-NAS	.922 [.910, .931]	.740 [.692, .779]

Table 18. **Recent common-population controls.** Both tested methods remain positive on the two NB201 axes.

J.1. Measurement provenance and compute accounting

Table 19 separates measurements obtained from public artifacts, statistics computed from those measurements, and measured at-init scores. In particular, the NBSZ-comparable ridge result is only one row of the analysis; the early-training target, aligned-axis residual analyses, spectral and probe features, transport diagnostics, and recent-proxy controls form distinct measurement blocks.

For compute accounting, the cached-statistics arms (cross-validation, correlations, bootstraps, orientation/sampling/dimension/external-coverage controls, and the search simulation) perform no network training. The timestamped recent-proxy RTX 4090 log sums to 3,444 seconds for SWAP and 365 seconds for AZ-NAS across 1,425 architectures. The frozen-NTK arm records 324×5 at-init Jacobian/Gram batches. The older spectral/probe logs record the device and exact architecture counts but not reliable start/end timestamps; we therefore report their work units rather than reconstructing an unsupported aggregate GPU-hour total.

J.2. Datasheet for the portability-map artifact

This datasheet documents the reusable cross-space portability map of Sec. 5.1; the body section (Sec. 5.1) carries the one-paragraph summary, while the population-reconciliation table (Tab. 20) and the heavy detail live here.

Motivation and intended use. The artifact records retrospective sign agreement for measured space pairs and supports auditing which proxy transfers deserve fresh target calibration. It is *not* an input to the evaluated search screen, a label-free accuracy predictor, or a substitute for measuring a proxy on a new space. In the search experiment of Sec. 5, the target sign is estimated anew on each split's target-CALIB cached accuracy and evaluated on target-EVAL.

Fields. The map is the 13×7 matrix of Spearman corr(Q, acc) (proxy $\times$ benchmark/task column) shown in

Measurement block	NB201 TSS/SSS	NB101 / NB301	TransNAS	Status and operation
Final performance, params, FLOPs	official benchmark caches	official benchmark caches	official benchmark caches	cached; no architecture training
Standard common proxies	cached common scores; nwot, l2-norm, and epe-nas separately scored on single-architecture nets	cached NBSZ-format proxy records	cached NBSZ-format proxy records	cached plus at-init scoring; no training
Early-training target T	computed from cached training curves	exact early T unavailable	exact T/final accuracy on four accuracy tasks	curve-derived measurement; no architecture training
Spectral and Q-probe zoo	scored on 500 architectures per axis, then intersected	unavailable	unavailable	at-init forward/backward statistics; no training
Frozen-NTK target	324 TSS architectures, five fixed batches	unavailable	unavailable	at-init Jacobian/Gram computation; no optimizer steps
PCA, residual/portability, and robustness controls	computed from aligned values	computed where required fields exist	computed where required fields exist	statistical analysis/resampling only; external Eq. (1) remained ineligible
SWAP and AZ-NAS controls	scored on 1,425 common TSS/SSS architectures	unavailable	unavailable	2,850 architecture-method rows (52,725 raw rows); no training

Table 19. **Measurement provenance and coverage.** "Unavailable" means that the aligned official artifact lacks the field; it is not imputed. All neural-network measurements performed in this work are at initialization rather than candidate training.

Table 20. **Population reconciliation for the headline correlations**: each headline corr(Q, acc) mapped to the population it was measured on, with its positive $(+)$ and flipped $(-)$ side; the numbers are the body's, reproduced verbatim, so a reader can match every figure to its n. The first five rows are the two NB201 axes (TSS/SSS); the jacob-cov row is a cross-space pair (TransNAS-scene vs. NB101).

Proxy	Population	$+$ side	$-$ side
cond-sum	full-zoo (n=324/138)	$+0.88$	-0.77
cond-sum	spectral arm (n=500)	$+0.887$	-0.781
meco	full-zoo (n=324/138)	$+0.88$	-0.79
meco	search population	$+0.882$	-0.811
grasp	full-zoo	$+0.55$	-0.19
jacob-cov	cross-space	$+0.78$	-0.53

Fig. 3. Table 5 tabulates exact point estimates and sample-size exceptions; intervals for the four strict cases appear in Appendix G.3. The logged per-architecture values retain proxy identifier, benchmark/task identifier, accuracy, and proxy score, so every point estimate, sample count, and interval can be re-derived.

Sign-stability annotation. For a measured (proxy, space-pair), the retrospective flag records whether sign corr(Q, acc) agrees across the full observed pair. This is a descriptive map annotation, distinct from the split-specific target-CALIB screen of Sec. 5. Degenerate or small-n cells (acc$\leq$10% dead architectures; NB301 n=50, disjoint) are flagged as suggestive, not load-bearing, matching the caveats of Appendix H.5.

Format and splits. The planned release is a tabular file keyed by (proxy, benchmark/task) with the fields above, plus the per-architecture value tables that back it. There is no train/test *learning* split: the artifact is a measurement table, not a trained model. The evaluated pre-screen instead reads source proxy values and a disjoint target calibration split of cached accuracy (Appendix I.4); it is target-EVAL-disjoint but not label-free.

Provenance. The cells are computed from the same per-architecture proxy and accuracy values used in the body (Appendices B and J); intervals use the bootstrap protocol in Appendix G.3. Populations are reconciled in Tab. 20 and detailed in Appendix H.1.

Limitations. The map is observational (Spearman correlations on cached accuracy and at-init proxies; no causal claim). Coverage is uneven: the deep capacity/trainability

decomposition is NB201-only, and the spectral proxies exist for NB201 only. The cross-space arm reintroduces dataset and n confounds that the within-NB201 design removes (Appendix H.5). The sign-stability flag is a binary sign-agreement summary, not a calibrated transfer-quality score, and the SSS arm is the weaker one (n=138 full-zoo). The pre-screen evaluation contains two evaluated trap directions, two null directions, K=1/5/10/50, standard baselines, and an identity-disjoint sensitivity. It remains target-calibrated rather than fully label-free and does not dominate every baseline (Appendix I.4).

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